

Labo 4 – Solution

204.1 Signal and information

Statistical signal analysis

Run the script `lab04-solution.jl` to see the solutions. The data were taken from year 1988.

Exercise 1

1. ~

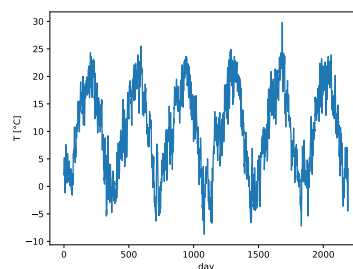


Figure 1: Daily average temperature.

2. ~

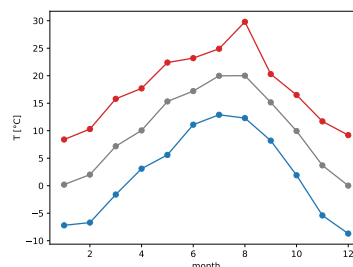


Figure 2: Monthly average, min and max temperature.

3. The signal is not stationary, mean changes over time, even though variance does not change much.

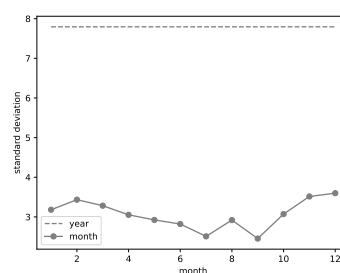


Figure 3: Monthly and yearly standard deviation of the temperatures.

4. Here, a good choice of bins is one bin per degree, e.g., each bin from $X.5^\circ\text{C}$ to $X.5 + 1.0^\circ\text{C}$.

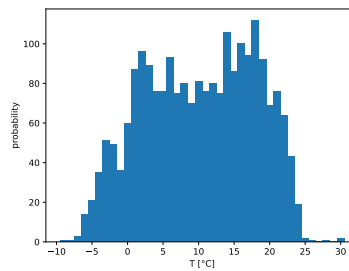


Figure 4: Histogram of the daily average temperatures for Sion in 1988.

5. The entropy is $H(T) = 4.88$.

Exercise 2

1. The correlation between average daily temperatures in Sion and Miami, between 1988 and 1993 is $r \approx 0.68$.
2. The temperatures are relatively well-correlated ($r > 0.5$), but likely to be taken in regions with different climates.
3. Autocorrelation is strong after one year, because temperature patterns repeat at this period.

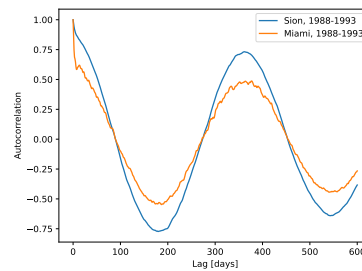


Figure 5: Autocorrelation over year 1988 and early 1989 for Sion and Miami.

Exercise 3

1. We observe outliers in Jan, Jun, and Dec for Sion, and in Jun, Aug, Oct, and Dec for Miami.

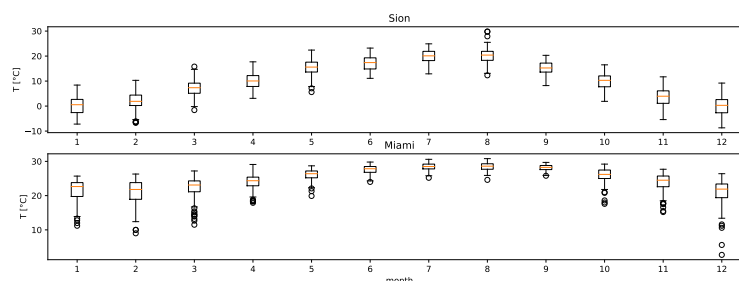


Figure 6: Monthly boxplot of daily average temperatures for Sion (top) and Miami (bottom).

2. The z -score is larger than 2 on 14.1, 15.1, 10.2, 22.7, 23.7, and 5.12 in Sion, and on 2.10 and 3.10 in Miami. These outliers do not match the ones in the boxplots.
3. For winter months, the spread is very large in Miami, but not in Sion. In summer, the spread is larger in Sion than in Miami (e.g., in August).

Exercise 4

Collect the data from Zurich, Geneva, Basel, Lausanne, Bern, Winterthur, Lucerne, St. Gallen, Lugano, and Bienne within the same dataset and perform the same analysis as before.